

better scores, etc. But what about teachers? What benefits do teachers get from AI technology? Will AI replace teachers in the near future? Before answering all these questions, let's first go through some facts about AI and its implications in the education industry.

This article covers some common AI-based solutions in education, use cases of AI-based learning platforms, and advantages of AI in the Indian education system.

AI-based solutions in education

There are many AI-based solutions available in the education field. There are great solutions available for school and college students. AI-based applications can analyse enormous amount of information and offer various options for students, including personalised learning materials.

Personalised learning. This is among the most important applications of AI that allows focusing on the individual needs of the student. There are firms like Dale Carnegie Training that offer many personalised courses. Through the use of AI technology, it is possible to create individual instructions, testing, and feedback.

Voice assistants. Some AI-based voice assistants available in the market include Amazon Alexa, Apple Siri, and Google Home. These gadgets are already in use for teaching and learning. Students can interact with these gadgets and get help without a human teacher. Since such AI-based assistants are new, most students find these quite interesting, exciting, and more engaging.

Smart content. This includes various learning materials, such as digitised textbooks, personalised contents, and customised interfaces.

Global learning. With AI technology, students can study various courses and get training programmes available in different parts of the world by sitting at home. There are many platforms with interactive learning materials available from the best tutors across the globe. AI also provides opportunities for students who speak different languages or have visual or hearing problems. There is an AI-based translator solution that creates subtitles in real-time mode. Using AI-based speech recognition, students can listen in their native languages.

Online classroom. Due to the out-



Virtual reality in education (Credit: xd.adobe.com)

break of coronavirus, online classes have become popular than ever across the globe. A lot of companies have launched online classroom apps where students and teachers can interact through live video streaming, screen sharing, media-interactive presentations, etc. The most common online classroom apps used in schools include Zoom, Microsoft Teams, and Google Meet. Zoom has rolled out AI-powered transcripts, note-taking features, facial recognition, and more. Microsoft Teams is another videoconferencing app with a rich set of AI-enabled capabilities. Google Meet is a popular app with AI-based noise cancellation for video calls.

Eyeball tracker. This tool tracks learners' eye gaze and provides real-time feedback on their gaze patterns during lectures. The main goal is to investigate the attention patterns of students in the classroom and create a better understanding of their interests. It also aims to generate new insights into effective teaching techniques and keep students engaged during lectures.

Emotion recognition. Some schools have equipped several classrooms with cameras to recognise emotions of students. The systems automatically take attendance and track the activities of students, including reading, writing, or listening. Emotion recognition uses images rather than text as inputs. The images could be static photos of students' faces taken in real time. Using the system, the school can observe the behaviour of each student and take actions accordingly.

Virtual reality (VR). VR with AI provides outstanding visualisations

that are not possible in traditional classrooms. The VR system uses immersive 3D visualisation technology to improve education and experiences. For example, when a student uses VR to understand the human heart, he/she will be able to visualise the whole system, including internal elements, pumping of blood back and forth inside the heart, and other functions. It helps in a better understanding of the concept.

AI-based learning platforms

There are many AI-based learning platforms already available across the globe. Some features of these platforms include the ability to analyse the knowledge level of a learner, offer backward communication, provide a plan for improvements, among others.

Beable. It is an integrated system launched by Achieve3000. It is a remote learning solution for middle school and high school in the US. It combines with data science, automation, and AI to deliver a tailored and multi-dimensional solution for students.

Little Dragon. It employs behavioural and emotional aspects of learning. An emotional AI detects states like boredom and frustration and then adjusts content to optimise the experience of learning.

Brainly. It is an AI-based solution offering personalised materials. It is basically a social learning community to help students. They can discuss issues related to their homework or gain new knowledge from other students. This platform utilises ML to provide a better user experience and assists in identifying spam and inappropriate content.